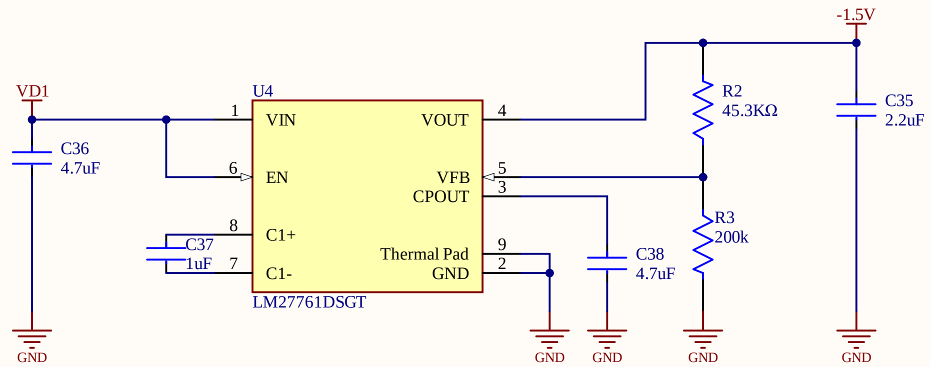
- Battery (VBAT)
- Always on 1.8V up to 750mA (1V8)
- System voltage between VBAT and VUSB (Vsys)
- Three software-controlled digital supplies (VD0, VD1, VD2)
- Two software-controlled analog supplies (VA0, VA1)

Default
- VD0: Off
- VD1: 1.8V or higher (Buck-Boost)
- VD2: Off
- VA0: Off
- VA1: 1.5 or 3.0V depending on config
- 1V8: Always On



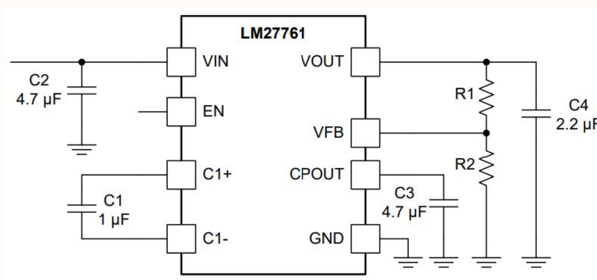


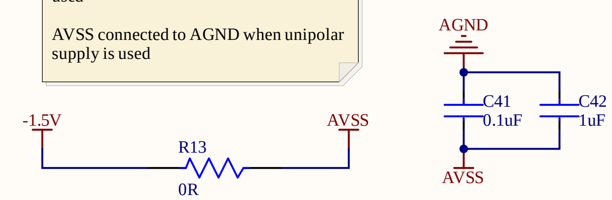
Figure 16. LM27761 Typical Application

$V_{OUT} = -1.22\ V \times (R1 + R2) / R2$

Analog supply for the ADS can be switched on/off software controlled with the PMIC.

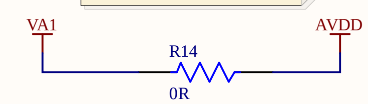
AVSS = -1.5V when bipolar supply is used

AVSS connected to AGND when unipolar supply is used

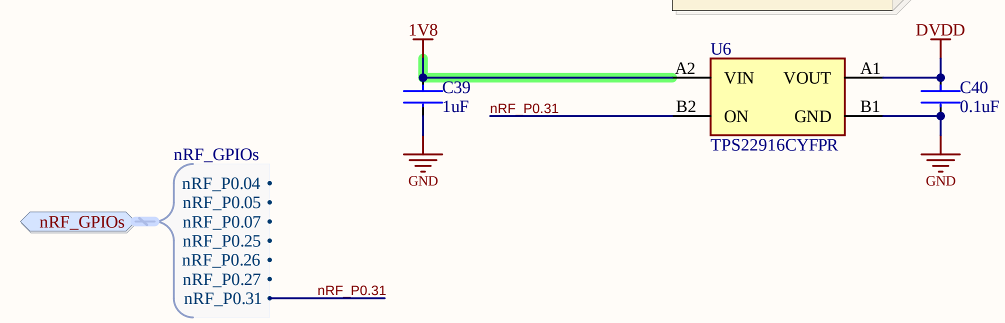


VA1 = 1.5V when bipolar supply is used

VA1 = 3.0V when unipolar supply is used



Digital supply for the ADS can be switched on/off with a load switch.



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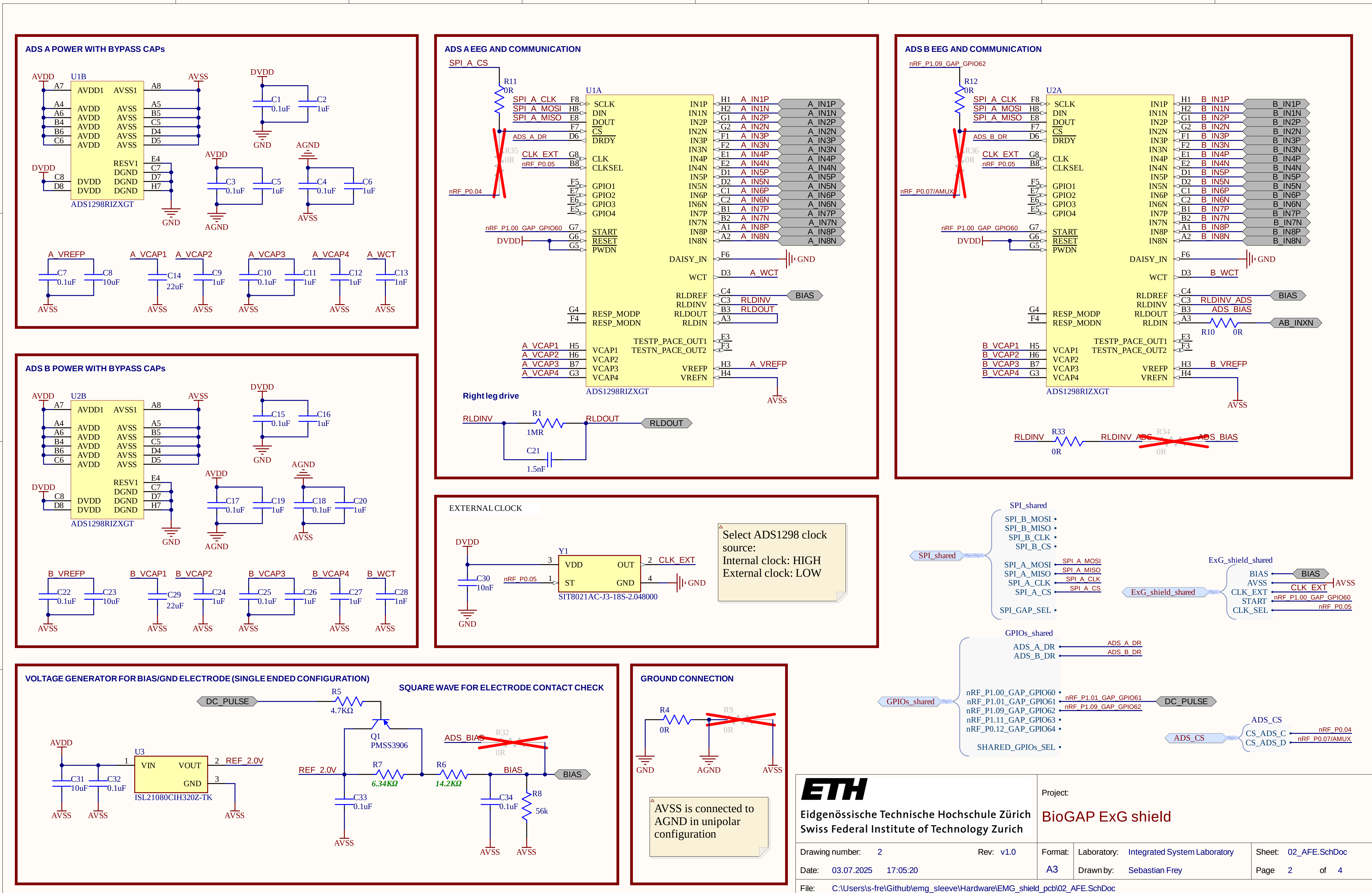
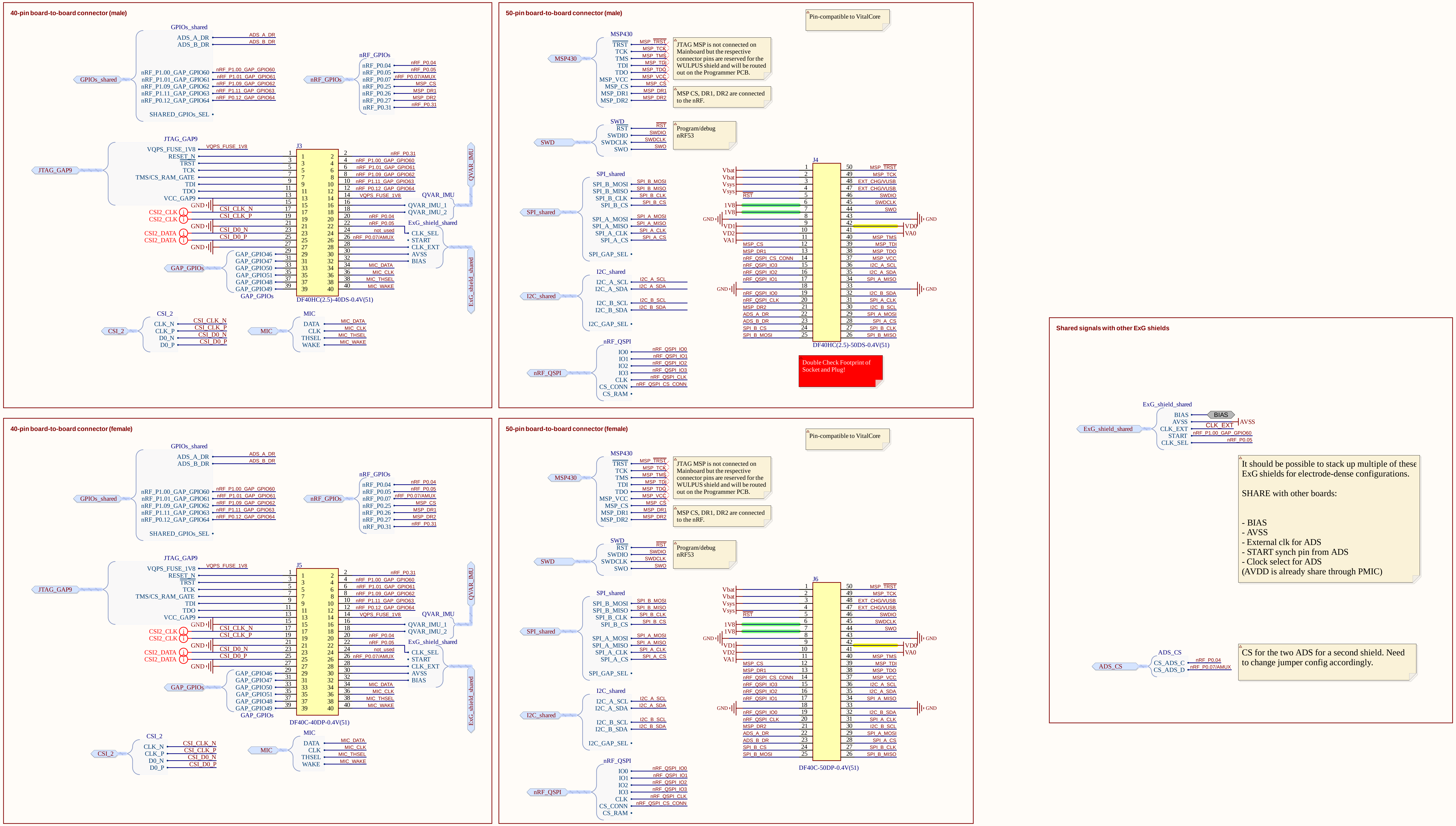
Project: BioGAP ExG shield

Drawing number: 1	Rev: v1.0	Format: A3	Laboratory: Integrated System Laboratory	Sheet: 01_power.SchDoc
Date: 03.07.2025 17:05:20			Drawn by: Sebastian Frey	Page 1 of 4

File: C:\Users\s-fre\Github\emg_sleeve\Hardware\EMG_shield_pcb\01_power.SchDoc


ADS A POWER WITH BYPASS CAPs
ADS A EEG AND COMMUNICATION
ADS B EEG AND COMMUNICATION
ADS B POWER WITH BYPASS CAPs
EXTERNAL CLOCK
Select ADS1298 clock source:
Internal clock: HIGH
External clock: LOW
Right leg drive
VOLTAGE GENERATOR FOR BIAS/GND ELECTRODE (SINGLE ENDED CONFIGURATION)
SQUARE WAVE FOR ELECTRODE CONTACT CHECK
GROUND CONNECTION
AVSS is connected to AGND in unipolar configuration
SPI_shared
ExG_shield_shared
GPIOs_shared
ADS_CS
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich
Project:
BioGAP ExG shield
Drawing number: 2
Rev: v1.0
Format: A3
Laboratory: Integrated System Laboratory
Sheet: 02_AFE.SchDoc
Date: 03.07.2025 17:05:20
Drawn by: Sebastian Frey
Page 2 of 4
File: C:\Users\s-fre\Github\emg_sleeve\Hardware\EMG_shield_pcb\02_AFE.SchDoc



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Drawing title: BioGAP ExG shield

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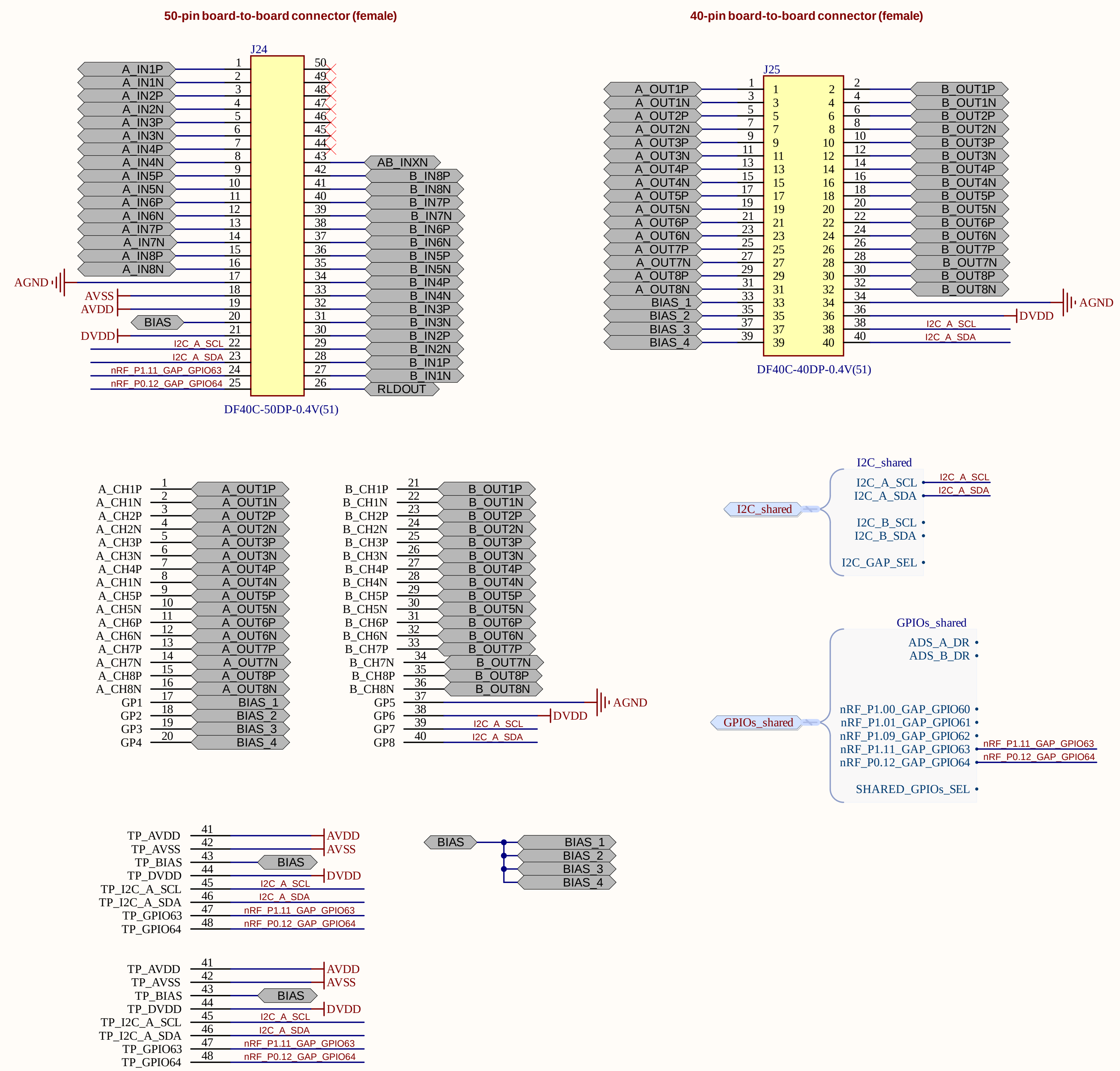
Date: 03.07.2025 17:05:20		Drawn by: Sebastian Frey	Project: 04_gap_interface.SchDoc
			Page 4 of 4

File: C:\Users\s-fre\Github\emg_sleeve\Hardware\EMG_shield_pcb\04_gap_interface.SchDoc

Board-to-board connector



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Project: BioGAP EMG shield

Drawing number: *	Rev: v2.0	Format: A3	Laboratory: *
Date: 03.07.2025 17:05:21			Drawn by: *

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Page * of *

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